

Rapid Spanning Tree Protocol (RSTP)

Rapid Spanning Tree Protocol (RSTP) is an improved version of the traditional Spanning Tree Protocol (STP) that prevents Layer 2 switching loops while allowing the network to recover from link failures much faster.

- STP: IEEE 802.1D
- RSTP: IEEE 802.1w
- Main purpose: Prevent switching loops and provide faster convergence.

1. Why do we need RSTP?

Consider three switches connected in a triangle:



Without spanning tree protection, Ethernet frames can circulate endlessly around the triangle.

RSTP logically blocks a redundant path while keeping it available as a backup. If the active path fails, RSTP can rapidly activate the backup.

2. STP vs. RSTP

Convergence time

RSTP is faster

30–50 s

Typical STP

Traditional STP may take tens of seconds to reconverge.

~1 s

RSTP

RSTP can often recover within a few seconds or less, depending on the topology.

The traditional STP convergence time is commonly described as approximately 30–50 seconds under its default timers. RSTP does not depend on waiting for the full STP timer sequence in normal point-to-point failures.

3. RSTP port roles

RSTP retains the familiar STP port roles, but adds a new backup role.

Root Port (RP)

The best path from a non-root switch toward the root bridge. Each non-root switch normally has one root port.

Designated Port (DP)

The forwarding port selected for a network segment. It provides the best path toward the root bridge for that segment.

Alternate Port

A backup path toward the root bridge. It is normally discarding but can quickly become the root port if the current root path fails.

Backup Port

A redundant path to the same shared LAN segment where another port on the same switch is designated. It is normally discarding.

Key difference: STP usually describes redundant ports as non-designated ports, while RSTP distinguishes between alternate and backup ports.

4. RSTP port states

Traditional STP	RSTP

Blocking	Discarding
Listening	Discarding
Learning	Learning
Forwarding	Forwarding
Disabled	Discarding or administratively down

What each RSTP state does

Discarding

Does not forward normal data frames or learn MAC addresses. It can still process BPDUs.

Learning

Learns source MAC addresses but does not yet forward normal data frames.

Forwarding

Forwards normal data frames and learns MAC addresses.

RSTP does not need separate Listening and Blocking states.

5. How RSTP achieves faster convergence

A. Proposal and Agreement mechanism

This is one of the most important RSTP improvements.

Imagine S1 is the root bridge, and S2 is connected to S1.

1. S1 sends a BPDU with a proposal that its connected port should be the designated forwarding path.
2. S2 checks whether it can safely accept the proposed path.
3. S2 temporarily synchronizes its other affected ports, placing them in discarding.
4. S2 sends an agreement BPDU back to S1.
5. S1 can immediately move the link to forwarding.

This handshake avoids waiting through the long traditional STP timers on suitable point-to-point links.

B. RSTP uses BPDUs differently

Feature	STP	RSTP
BPDU transmission	Root bridge originates configuration BPDUs; other switches relay them	Every switch generates BPDUs at regular intervals
BPDU version	Version 0	Version 2
Failure detection	Often relies on Max Age expiration	Can detect missed BPDUs much faster
Recovery	Timer-based in many cases	Proposal/agreement and alternate paths

Important: RSTP still uses the Hello timer, normally 2 seconds. A switch can declare a neighbor disconnected after missing three consecutive BPDUs, giving a common default detection time of about 6 seconds. Some failures can be recovered even faster through the RSTP handshake.

6. RSTP link types

RSTP classifies links to determine how quickly they can transition.

Point-to-point

A full-duplex link between two switches. RSTP can use proposal and agreement for rapid transition.

Fast convergence possible

Shared

A shared-media link, traditionally associated with half-duplex Ethernet. It uses a more conservative transition process.

Slower than point-to-point

Edge port

A port connected to an end device such as a PC or printer, not another switch. It can move directly to forwarding.

Equivalent to PortFast behavior

7. Edge ports and BPDU protection

In RSTP, an edge port is similar to Cisco's PortFast-enabled port.

- It immediately enters forwarding when the link comes up.
- It is intended for end devices.
- If it receives a BPDU, it is no longer considered a safe edge connection.
- With BPDU Guard configured, receiving a BPDU can place the port into an error-disabled state.

Do not configure edge ports on links between switches.

8. RSTP topology example

Suppose S1 is the root bridge.



Green: Forwarding

Blue: Forwarding

Orange: Discarding backup

If the S1–S2 link fails

1. S2 detects that its root port has failed.
2. S2 already has an alternate path through S3.
3. RSTP can select that alternate port as the new root port.
4. The alternate port transitions toward forwarding, restoring connectivity.

This is much faster than relying on the traditional STP timer sequence.

9. RSTP configuration on Cisco switches

On Cisco IOS, Rapid PVST+ is commonly configured using:

cisco

Switch(config)# spanning-tree mode rapid-pvst

To configure a primary root bridge for a VLAN:

cisco

Switch(config)# spanning-tree vlan 10 root primary

To configure a secondary root bridge:

cisco

Switch(config)# spanning-tree vlan 10 root secondary

To verify:

cisco

Switch# show spanning-tree

To inspect the RSTP mode and port roles:

cisco

Switch# show spanning-tree summary

Note: **rapid-pvst** is Cisco's per-VLAN implementation of RSTP. It is not exactly the same as the standards-based single-instance RSTP mode.

10. Quick revision

Remember these points

Exam revision

- RSTP = IEEE 802.1w.
- It prevents Layer 2 loops and converges faster than traditional STP.
- RSTP has three port states: Discarding, Learning, Forwarding.
- RSTP adds Alternate and Backup port roles.
- Proposal and Agreement enable rapid transitions.
- Edge ports are the RSTP equivalent of PortFast behavior.
- RSTP still uses BPDUs and elects a root bridge.
- Cisco Rapid PVST+ provides rapid spanning tree per VLAN.

Test your understanding